The Topology of Reachable Nature: Persistent Homology and the Resilience of Green-Space Access in Cities of Contrasting Morphology

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Abstract

Sustainable Development Goal 11.7 calls for universal access to green and public space, yet accessibility is almost always quantified as a static snapshot that says nothing about how access degrades when the street network is disrupted. A *topological resilience metric* for accessibility is introduced: the Wasserstein distance between the persistent-homology diagram of a city's green-space accessibility field before and after network disruption, traced as a function of disruption intensity ρ. The area under this degradation curve and its critical transition ρ^* summarise how robustly a place keeps its *locations* close to nature. This is one facet of SDG 11.7, the pedestrian network's reach to green space, and not its population-coverage dimension. Applying the metric to eight morphologically contrasting cities on four continents (İstanbul, Barcelona, Amsterdam, Bogotá, Phoenix, Singapore, Nairobi, Vienna) at the level of 3,999 intra-urban districts, street-network morphology is found to be consistently associated with the resilience of the *connected* green-space access structure. In a district-level regression with city fixed effects, longer street segments (the largest effect) and more circuitous routing are associated with greater resilience and higher orientation entropy with less; these are unstandardised associations, and standardised and percentile effect sizes are reported because the sample is large. Circuity and mean street length are the robust pair—significant under a wild cluster bootstrap (the appropriate small-cluster correction for the eight cities) and under spatial-error and spatial-lag models—while orientation entropy is significant and strong in the spatial models but less certain across cities. The signs are stable across random and targeted disruption and district resolutions spanning a 28× range. The metric itself passes a within-city criterion check: it correlates positively with a distance-based accessibility-degradation benchmark (Spearman 0.36) and, using the open GHS-POP grid, with a population-weighted one (0.35), while adding information beyond both, with the scope that it captures the resilience of the *connected* served structure and not total disconnection. The signal lives in dimension-0 homology, the merging of well-served regions, and not in enclosed "access deserts," as observed in the real street networks examined here.

1 Introduction

Target 11.7 of Sustainable Development Goal 11 commits states to provide "universal access to safe, inclusive and accessible, green and public spaces" [18]. Planners typically measure progress with catchment or proximity analyses: the share of residents within a walking threshold of a park. Such measures are static. They describe access on an ordinary day and are silent about *resilience*—how access to nature holds up when the pedestrian network is degraded by a flood, an earthquake, or the closure of a critical bridge or street. Two gaps follow. First, a city can score well on average proximity yet collapse under modest disruption; average accessibility does not reveal this fragility. Second, accessibility work rarely connects the *shape* of access to urban *form*, which is precisely the architecture–mathematics question that motivates this collection.

Persistent homology (PH) offers a principled, multi-scale, coordinate-free language for the "holes" and connected pieces of a coverage pattern [9, 13]. PH has been used to detect coverage gaps for amenities such as polling sites [11] and healthcare [10], to read the topology of urban congestion [19], and to quantify displacement under urban expansion [14]. What is *not* novel is using PH to find accessibility "holes" for an amenity. The contribution here is to move from *detecting* coverage gaps to *measuring the resilience of the access topology* and to *linking that resilience to urban form.* A scope note is in order at the outset: what is measured is *network* accessibility, the pedestrian graph's spatial reach to green space with every network node weighted equally, and not *population* accessibility, which would weight locations by the residents they serve. The two coincide only where population is uniform, so the claims made here concern the network's structural capacity to keep places near nature, not how many people retain access.

Contributions.

- **A topological resilience metric** (the mathematics). $D(\rho)$ is defined as the Wasserstein distance between the persistent-homology diagram of the green-space accessibility field before and after disruption of intensity ρ, and is summarised by an area-under-curve (AUC) and a critical transition ρ^* (Section 4).
- **First topological treatment of SDG 11.7 green-space access** in an architecture–mathematics framing.
- **A district-level morphology↔resilience result** across 3,999 districts in eight cities, with city fixed effects, showing a consistent association between street-network morphology and access resilience whose sign is stable across disruption models and district resolutions (Section 6).

2 Related work

TDA of spatial systems. Feng et al. [9] survey persistent homology for spatial data; Otter et al. [13] give the computational foundations. Within urban analysis, Wu et al. [19] read traffic congestion as a barcode, and a line of "resource coverage" work applies sublevel or network-distance filtrations to amenity access: polling places [11] and sexual-health services [10]. Rodríguez Vázquez and Cuerno [14] use PH to quantify displacement under urban expansion. These studies establish that PH can *locate* coverage holes. The object of study here differs: not where the deserts are, but how the whole access topology *degrades* under disruption, with that degradation regressed on morphology.

Accessibility and equity. A large body of work measures green-space access as a static field, whether by simple footpath proximity [4] or by catchment methods such as the Gaussian two-step floating catchment area [16], often to diagnose inequities in who lives close to nature [1]. The accessibility field used here (Section 4) is built the same way (network walking time to the nearest green space) but is then read topologically and stress-tested, rather than thresholded once.

Street-network morphology. Form is characterised with standard descriptors computed from OpenStreetMap via OSMnx [2]: intersection density, circuity, orientation entropy, and mean street-segment length. Relating these to a resilience outcome is the architecture–mathematics bridge of this paper.

3 Mathematical background

Let $G = (V, E)$ be the pedestrian network and $f : V \to \mathbb{R}$ a scalar field giving, at each node, the network walking time to the nearest green-space access point. The *sublevel-set filtration* of f builds a growing simplicial complex $K_t = \{\, \sigma : \max_{v\in\sigma} f(v) \le t \,\}$. As the threshold t (walk-minutes) rises, well-served regions appear and merge. Persistent homology records, for each topological feature, a birth b and death d; the multiset of points (b, d) is the *persistence diagram* $\mathrm{Dgm}_k(f)$ in homological dimension k [8]. Dimension 0 tracks connected components (the merging of well-served regions); dimension 1 tracks loops (enclosed voids, i.e. access "deserts").

The distance between two diagrams is measured by an optimal matching. The 1-Wasserstein distance is

$$W_1(X, Y) = \min_{\gamma} \sum_{x\in X} \|x - \gamma(x)\|,$$

minimised over bijections γ between the diagrams (with matching to the diagonal allowed). For a *fixed* domain, these diagram distances are stable to perturbations of the field f: the bottleneck distance by the classical stability theorem [5], and the 1-Wasserstein distance used here by the L_p-stability result of Cohen-Steiner et al. [6]. A caveat must be stated precisely here, because it bounds what stability can justify. These theorems control the effect of changing the *field* on a fixed simplicial complex; the disruption in this study, however, also removes edges, changing the underlying graph (the domain) and not only f. The classical stability theorem therefore does *not* directly bound $D(\rho)$: it guarantees only that the two ingredients that *are* pure field changes—re-snapping the accessibility field on the surviving subgraph—behave continuously, not that the whole map from disruption intensity to diagram distance is well-behaved. In particular, stability does not imply monotonicity, and $D(\rho)$ is indeed *not* always monotone: at high ρ the İstanbul and Phoenix curves turn down as the network fragments and finite classes vanish (Section 6). Stability is thus a mathematical well-posedness property of the field-to-diagram map, not a guarantee about the metric under network damage; the metric is treated throughout as a *proposed proxy* for how the access topology holds up, to be validated against independent resilience evidence in future work rather than as ground truth. PH is computed with GUDHI [17] and diagram distances with `persim`.

4 Methodology

Data. For each city the pedestrian network and green spaces are taken from OpenStreetMap via OSMnx [2], and the analysis extent is fixed to the city's *urban centre* polygon from the GHS Urban Centre Database [12] so that extents are comparable across countries. Green spaces are the OSM classes `leisure=park|garden|recreation_ground`, `landuse=grass|recreation_ground`, `place=square` and `natural=wood`; their centroids, snapped to the nearest network node, are the access points. These are termed *OSM-mapped green* spaces, not *public* spaces: OSM tags do not certify public accessibility, so a private garden, a fenced lawn or an impenetrable wood can enter the set. The classes are included because together they capture the everyday green and open spaces a pedestrian can plausibly reach; the trade-off is heterogeneity, addressed by the class-exclusion sensitivity below. This is a deliberately inclusive definition: it merges spaces of very different type, size and usability (a pocket garden, a city square and a large wood all count equally), and it takes the centroid as a single access point rather than modelling entrances or area. A sensitivity analysis under a strict, accessibility-conservative subset (dropping wood, grass and gardens; keeping parks, recreation grounds and squares) is reported below. Every green space is treated as an equivalent point source; distinguishing quality, size, or public/private status is left to future work (see Limitations).

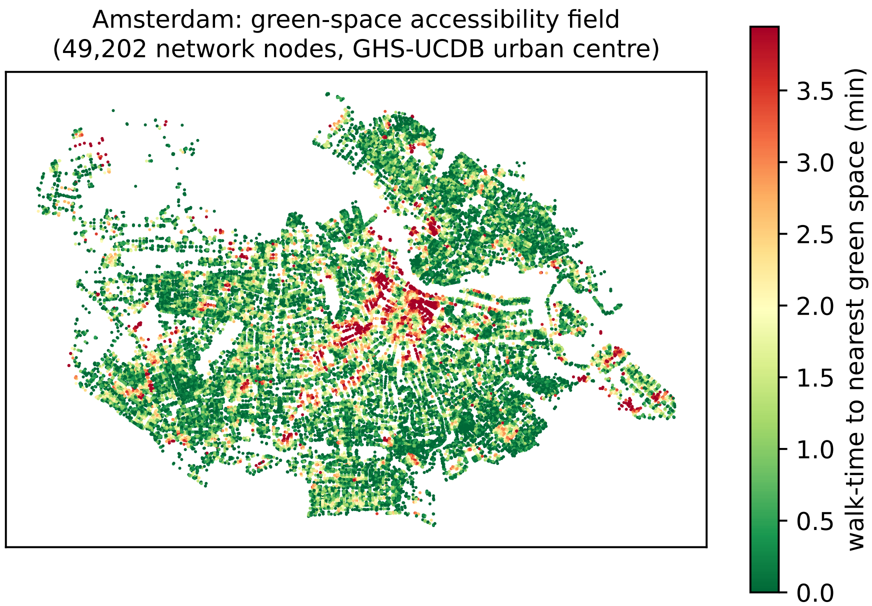

Figure 1: Green-space accessibility field for Amsterdam: each of the 49,202 pedestrian-network nodes inside the GHS-UCDB urban centre, coloured by network walking time to the nearest green space. Green marks well-served locations, red the access-poor pockets whose merging structure the resilience metric tracks.

Accessibility field. Each node is assigned its network walking time to the nearest green-space access point by multi-source Dijkstra, giving the field f (Fig. 1). Length is converted to time at a constant 4.8 km/h (1.33 m/s), which sits within the comfortable adult gait speed of 1.27–1.46 m/s reported by Bohannon [3]. The exact value does not matter for the conclusions: speed enters only through time = length/speed, so a different speed rescales the whole field, and hence every diagram, Wasserstein distance and AUC, by the same constant. The regression signs, significances and the city ranking are therefore invariant to the speed choice (if the persistence threshold is rescaled with speed the invariance is exact). This is confirmed empirically: on Amsterdam, recomputing at 4.0 and 5.0 km/h leaves the district AUC a near-exact rescaling of the 4.8 km/h values (Spearman $\rho = 0.98$ and 0.996; median AUC ratios 1.24 and 0.96, matching the expected 4.8/4.0 and 4.8/5.0). The field is defined over *network nodes*, each weighted equally; it is a property of the pedestrian network's spatial reach to green space, not a population-weighted or distributional measure. The metric therefore speaks to one facet of SDG 11.7, the network's capacity to keep locations close to green space and how that capacity degrades under disruption, and not to the target's population-coverage or inclusivity dimensions, which would require census and land-use data not used here.

Resilience metric. Let $\mathrm{Dgm}(f)$ be the baseline diagram. A disruption of intensity $\rho \in [0, 1]$ removes a fraction ρ of network edges; the field and diagram are recomputed on the disrupted network, defining the degradation

$$D(\rho) = \mathop{\mathbb{E}}_{r} W_1\big(\mathrm{Dgm}(f),\, \mathrm{Dgm}(f_{\rho,r})\big),$$

averaged over r replicate disruptions. Three disruption families are considered: uniformly random edge removal (percolation-style), targeted removal of high-betweenness edges, and hazard removal of edges in low-elevation zones. These are deliberately *stylised stress tests*, not simulations of specific disasters: random removal probes generic structural robustness (and, being spatially uniform, does *not* reproduce the spatially clustered damage of a real flood or earthquake), targeted removal probes the worst case of losing critical links, and the hazard proxy adds a first, mild topographic signal. They serve to compare places under a common, reproducible perturbation, and the results are read correspondingly cautiously as real-disaster resilience (Limitations). Exact edge betweenness is $O(V|E|)$ and intractable at city scale, so the targeted scenario ranks edges by betweenness approximated from $k = 500$ sampled pivot sources. The resilience of a place is summarised by the area under the curve $\text{AUC} = \int_0^{\rho_{\max}} D(\rho)\, d\rho$ (normalised by the ρ-range) and the critical transition ρ^*, the smallest ρ at which $D(\rho)$ first reaches half its maximum. The half-maximum serves as a scale-free, monotone summary of *where* the degradation curve turns over: unlike a fixed absolute threshold it adapts to each place's own range, and unlike the inflection point it is robust to the curve's noise. It is a descriptive locator of the transition, not a claim of a sharp percolation threshold, and (like AUC) it depends on the chosen disruption range $[0, \rho_{\max}]$; ρ^* is therefore compared only within a fixed range.

Homology dimension and denoising. On real street networks the sublevel filtration's finite H_1 is nearly empty: a street graph is close to planar and contains few triangles, so the network's cycles are never filled and almost all H_1 classes are essential (infinite) and carry no finite persistence. In Amsterdam, for example, the baseline diagram has 6,449 finite H_0 classes but only a single finite H_1 class (Fig. 2). This is a joint property of the *representation* (a node-valued walk-time field on the near-planar pedestrian graph) and the *sublevel filtration*, not a universal fact about green-space access. A different construction, for instance a Vietoris–Rips filtration on straight-line distances or a 2-complex that fills faces, would populate H_1 and could shift the signal into loops. Under this construction the finite information lives in H_0 (the merging of well-served regions), so $D(\rho)$ is computed on H_0 and the H_1 scarcity is treated as a modelling consequence to revisit with alternative filtrations, not as evidence that access "deserts" are absent. To keep the exact Wasserstein matching tractable on city-scale diagrams, H_0 classes with lifetime below 1 walk-minute are discarded. This cut is principled rather than arbitrary: a one-minute difference in walk time is below the resolution at which the constant-speed (4.8 km/h) field is meaningful, so sub-minute classes are measurement noise, not access structure. Its effect is small and was tested directly (Persistence-threshold sensitivity, below): varying the cut over 0.5–2.0 minutes preserves every sign and the significance of the three headline descriptors, changing only reported magnitudes.

Districts and inference. Statistical inference rests not on eight cities but on intra-urban *districts*: each urban-centre boundary is tiled with a uniform H3 hexagonal grid (resolution 8, $\approx$ 0.7 km^2 cells) and, per district, its local resilience summaries and its local morphology are computed (intersection density, circuity, orientation entropy, mean segment length, and a green-space fragmentation index—the number of distinct green-space patches per square kilometre of green area within the district, computed on the green space clipped to each hex so that it varies between districts rather than being a per-city constant collinear with the fixed effects). The following model is then fit

$$\text{AUC}_i = \beta^\top \mathbf{m}_i + \alpha_{c(i)} + \varepsilon_i,$$

an ordinary-least-squares regression of district AUC on morphology $\mathbf{m}_i$ with *city fixed effects* $\alpha_{c(i)}$ absorbing all *additive between-city* differences (any confound constant within a city, such as climate, planning regime or data vintage). They do not remove within-city confounding or effects that vary across districts inside a city; those are addressed separately with explicit controls and spatial

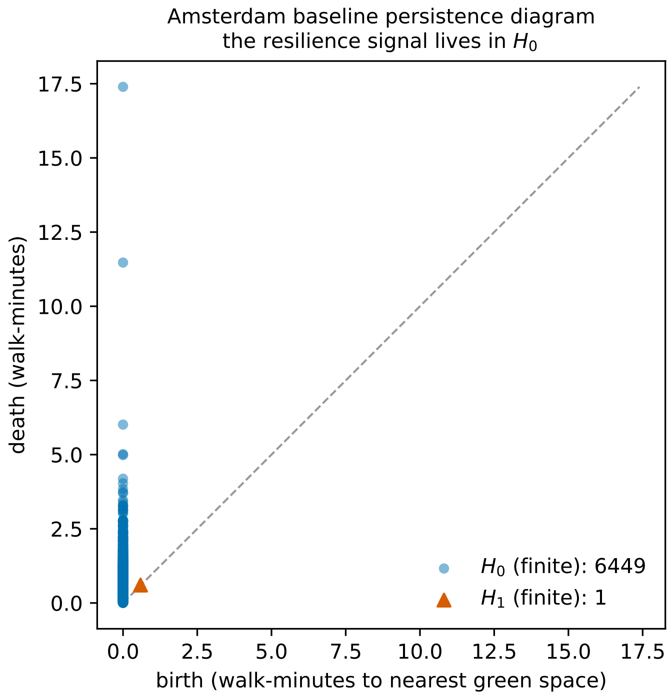

Figure 2: Persistence diagram of the Amsterdam baseline accessibility field. The finite part is carried almost entirely by H_0 (6,449 classes—the merging of well-served regions as the walk-time threshold rises) with only a single finite H_1 class, motivating an H_0-based resilience metric.

models below. The cities serve as a typological overlay, not as the inferential sample. This design also sidesteps the cross-city comparability problem of AUC. Because $\alpha_{c(i)}$ absorbs each city's overall AUC level, which scales with network and diagram size, the morphology coefficients are identified purely from *within-city* variation, and the inference never rests on comparing absolute AUC across cities. Only the descriptive city typology, not the regression, would need a size normalisation. For inference that respects the nesting, standard errors are also clustered at the *city* level—the eight clusters are the eight cities, the level at which each disruption and each fixed effect operate, and within which districts share a network, climate and planning regime. Eight clusters is a small number, at which the asymptotic cluster-robust p-values are only approximate; they are therefore reported alongside a wild cluster bootstrap (Appendix A), and the cities are treated as the practical, if not strictly independent, units of between-city variation.

5 Case studies

Eight cities on four continents are studied, chosen to span street morphology, planning regime and the Global North/South divide: İstanbul (organic, hilly, polycentric, seismic), Barcelona (Cerdà grid with superblock interventions), Amsterdam (compact, canal, walkable), Bogotá (dense informal–formal mix), Phoenix (low-density grid sprawl), Singapore (planned, tropical), Nairobi (rapidly growing, mixed-formality) and Vienna (compact, central European on the Danube). Boundaries are the GHS-UCDB urban-centre polygons; homonymous centres (e.g. Barcelona, Spain vs. Venezuela) are disambiguated by largest true (equal-area) extent. Two further candidates (Copenhagen, Melbourne) were dropped because the OpenStreetMap place query did not resolve their full urban

Table 1: District coverage and city typology (descriptive; H3 resolution 8, n = 3,999 across eight cities). Higher mean AUC $\Rightarrow$ access topology changes more under disruption $\Rightarrow$ *less* resilient.

City	districts	mean AUC	mean ρ^*	mean H_0 tot. persist.
Amsterdam	227	4.46	0.25	13.11
Nairobi	363	5.90	0.11	9.09
Vienna	460	6.08	0.22	12.91
Phoenix	962	6.68	0.16	8.08
Singapore	573	7.34	0.17	11.96
Barcelona	147	7.67	0.28	13.63
İstanbul	729	7.75	0.20	12.84
Bogotá	538	8.02	0.24	16.90

extent. Some robustness and validation analyses (Section 6, Appendix A) were computed on the original five-city core and are labelled as such.

6 Results

The random-disruption analysis yields 3,999 qualifying districts (147–962 per city, Table 1); only 2.6% have zero AUC. District mean AUC ranks Amsterdam as the most resilient and İstanbul/Bogotá as the least; Nairobi has the earliest breakdown ($\rho^* = 0.11$). The district-mean and whole-city rankings agree at the extremes (Amsterdam most resilient, Bogotá least) but not in the middle—Barcelona is mid-ranked by district mean yet most resilient at the whole-city scale—a reminder that AUC is comparable across cities only in broad ordering, not level.

The headline result is the district-level regression (Table 2), fit over eight cities and the full disruption grid (8 intensities $\times$ 10 replicates). With city fixed effects and n = 3,999, three of five morphology descriptors are significant, all at $p < 10^{-16}$. Because the coefficient sign is on AUC (higher AUC = less resilient), a negative coefficient means the descriptor is associated with *greater* resilience. More circuitous networks (coef −7.39) and longer street segments (−0.050) are associated with more resilient green-space access, whereas higher orientation entropy (+1.85) is associated with less (Fig. 3). Because the raw coefficients sit on different scales, magnitude is read from the standardised β and the 10th–90th percentile effect (Table 2): *mean street length* is the largest effect (std. $\beta = -0.28$; a 10th-to-90th-percentile change moves AUC by 62% of its standard deviation), then orientation entropy (+0.16) and circuity (−0.13); the tiny p-values reflect the large sample, not effect magnitude. The descriptors are not collinear (all variance-inflation factors < 3), and the pooled signs are not a pooling artefact—each of the three holds in seven of the eight cities. Intersection density is not significant ($p = 0.71$) and its sign is unstable across samples, and green-space fragmentation is likewise not significant. Districts nest within the eight cities and are spatially contiguous, so their residuals are not independent; Appendix A treats this in full. Under a wild cluster bootstrap—the appropriate small-cluster correction for the eight clusters (the eight cities)—circuity and mean street length remain significant ($p = 0.0002$ each), while orientation entropy does not ($p = 0.16$); orientation entropy is nonetheless strong in the spatial-error and spatial-lag models ($|z| \approx 10$). Circuity and mean street length are therefore treated as the robust pair and orientation entropy as a real but less certain third effect.

The whole-city scale gives an *illustrative* typology, not a second inferential result: it carries no statistical weight, and it is reported to visualise the district-level signal, not to rank cities definitively.

Table 2: District fixed-effects regression, AUC $\sim$ morphology + C(city), n = 3,999 across eight cities, random-disruption scenario at the full disruption grid (8×10). The standardised coefficient (std. β) and the change in AUC across each descriptor's 10th–90th percentile (as % of the AUC standard deviation) so that effects on different scales are comparable and not read off the raw coefficient or the (large-n) p-value alone. See Table 3 for city-clustered and spatial-model inference. Sig.: $^{***}p < 0.001$.

Morphology descriptor	coef	std. β	$\Delta_{10\text{–}90}$ (% SD)	p-value
mean street length	-0.0505	-0.28	-62	4.9×10^{-29} ***
orientation entropy	$+1.845$	$+0.16$	$+51$	1.4×10^{-27} ***
circuity	-7.387	-0.13	-27	3.6×10^{-17} ***
intersection density	$+0.021$	$+0.01$	$+2$	0.71
green-space fragmentation	-3.7×10^{-7}	-0.00	-0	0.79

For the five original cities, computed on the same UCDB-clipped networks, the degradation curves $D(\rho)$ rise smoothly and order the cities consistently with the district means (Fig. 4). Because city-level AUC is computed on the full-city diagram (thousands of H_0 classes) it is two to three orders of magnitude larger than the per-district means of Table 1 and scales with network size; it is *not* normalised, so only the relative ordering—never the level—should be read, and a size-normalised cross-city AUC is left to future work. On that ordering, city-level AUC is lowest, and so most resilient, for Barcelona (620) and Amsterdam (898), and highest for İstanbul (2440) and Bogotá (2828), with Phoenix in between (2053). The cities that would informally be called compact, Barcelona and Amsterdam, retain more of their access topology under disruption than the large, sprawling or polycentric ones; this pattern is described with the measured morphology descriptors rather than with "walkability," which is not modelled directly. Under extreme removal ($\rho = 0.5$) the İstanbul and Phoenix curves turn down as the network fragments so heavily that unreachable nodes drop out of the finite diagram, a signature of near-total collapse rather than of recovered access.

Robustness (summary). The morphology relationship is stable across every alternative specification tried; Appendix A gives the full results (the additional sensitivity checks were run on the original five-city core and are labelled there) and Fig. 5 summarises them. Briefly: the three headline signs and their significance ($p < 10^{-3}$ or better) survive a *targeted* betweenness disruption, the coarse 5×3 and full 8×10 disruption grids, H3 resolutions 7/8/9 (a 28× range of district counts), persistence thresholds from 0.5 to 2.0 walk-minutes, a halving and doubling of the betweenness pivot count, and the inclusion of four district-level controls (green-space area, road hierarchy, relief and city-centre distance). Intersection density is the one descriptor that moves: significant under some specifications but not the full-grid headline, so it is treated as the weakest, grid-sensitive effect throughout. The one disruption family that does *not* carry the signal is the spatially clustered real-flood hazard, which is treated as a distinct question (below).

Green-space definition sensitivity (five-city core). Because the inclusive green-space set mixes reliably public parks with possibly-private gardens, fenced grass and impenetrable woods, the analysis was re-run on a strict, accessibility-conservative subset: only `leisure=park`, `recreation_ground` and `place=square`, dropping wood, grass and gardens. This is a large cut (in Amsterdam, from 23,657 features to 369), yet the district AUC is largely preserved (Spearman 0.82 against the inclusive set on the same reduced grid), and the regression is unchanged in substance: circuity

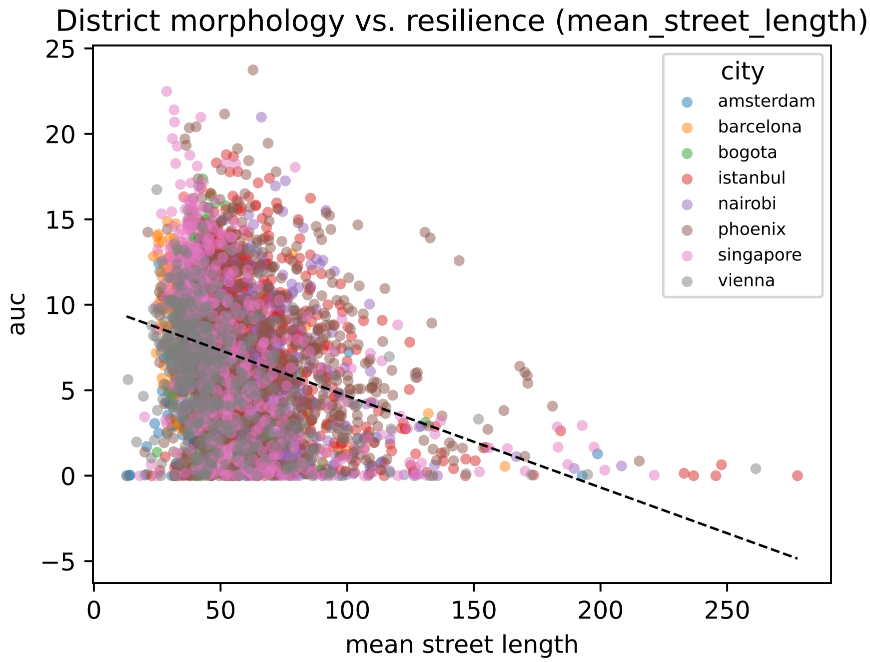

Figure 3: Headline morphology↔resilience relationship. Each point is one of the 3,999 districts across eight cities, coloured by city; the dashed line is the pooled least-squares fit. Higher AUC indicates a district whose green-space access topology degrades more under disruption (less resilient).

(-8.08, $p = 4 \times 10^{-9}$), mean street length (-0.048, $p = 2 \times 10^{-12}$) and orientation entropy ($+1.21$, $p = 2 \times 10^{-7}$) keep their sign and significance, and intersection density remains the weak, non-robust term. The morphology result is thus not an artefact of counting inaccessible green as accessible.

Criterion validity: benchmark against an accessibility measure (five-city core). If the topological machinery is to be worth its cost it should track a simpler measure of access loss. It is benchmarked, per district, against a *distance-based* accessibility-degradation measure, namely the increase in mean walk-time to green space under the same disruption (a proximity-accessibility degradation of the kind used throughout the field). As a second reference, a *connectivity* degradation, the increase in the fraction of nodes left unable to reach any green space; this is the natural non-persistent counterpart, since H_0 of a sublevel filtration is single-linkage connectivity. The level of comparison is what settles the question. The inference identifies morphology from *within-city* variation (city fixed effects), so the matching validity question is also within-city, once each city's overall AUC level (which scales with network size) is removed. Within city, the topological AUC correlates positively with the distance benchmark, with a *moderate* magnitude (within-city pooled Spearman $\rho = +0.36$; the tiny $p < 10^{-79}$ reflects the sample size, not strength). The correlation is positive in every city but heterogeneous, and weak in some: Amsterdam +0.47, Bogotá +0.16, İstanbul +0.10, Barcelona +0.05, Phoenix +0.03. This is moderate supporting evidence—not strong validation—that the metric tracks a simple accessibility-degradation ground truth at the level at which it is used. The naive pooled correlation is much weaker ($\rho = +0.07$) only because cross-city AUC scale swamps it, a Simpson's paradox driven by the same scale the fixed effects absorb. The benchmark also holds when access is *population-weighted* rather than node-uniform: weighting each node by the open GHS-POP residential-population grid [15] (JRC, 1 km, 2020; the same GHSL

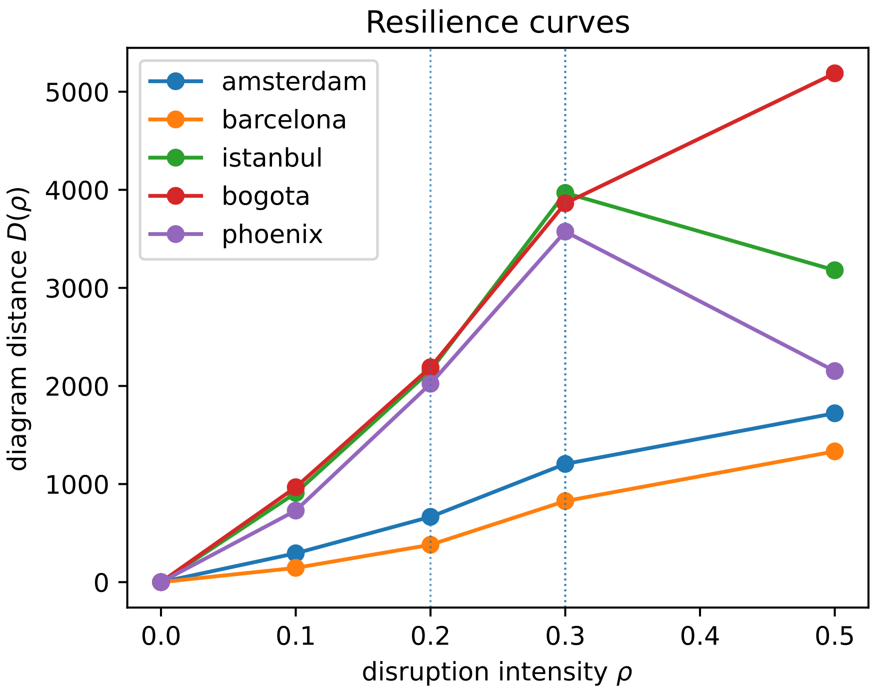

Figure 4: City-level resilience curves $D(\rho)$ for the five original cities (random-disruption scenario), computed on the UCDB-clipped networks; illustrative. Dotted vertical lines mark each city's critical transition ρ^*.

family and Mollweide grid as the urban-centre boundaries) gives an almost identical within-city correlation ($\rho = +0.35$, $p < 10^{-76}$). So the topological signal is not an artefact of weighting locations equally, and it speaks, at least for graded access, to the population-coverage dimension of SDG 11.7 and not only to bare network reach. Two qualifications remain. The metric adds value rather than echoing the benchmark: controlling for both baselines and city fixed effects, the three headline morphology descriptors stay significant ($p < 10^{-6}$), so it carries configurational information the simple measures miss. And the convergent validity is with *graded* access loss, not with *disconnection*. Within city, AUC correlates near zero-to-negative with connectivity degradation, because disconnected nodes leave the finite diagram. The metric is thus validly read as the resilience of the *connected* served structure, and should be paired with a disconnection count for total access loss; the Limitations show that neither a naive finite cap nor extended persistence fixes this cleanly (both recover the connectivity sign only by sacrificing the graded signal), so the two channels are kept separate and the claim is scoped accordingly.

7 Discussion

The results give an interpretable, architecture-facing reading of resilience. The robust effects are circuity and segment length: districts with more circuitous routing and longer segments retain more of their well-served structure when edges are removed, because alternative paths to green space survive the loss of any one link. Intersection density is not significant, and its sign is not even stable between the five- and eight-city samples, so no conclusion is drawn from it. Districts with more disordered street orientation (higher entropy) are more brittle. That the signal is carried by H_0 rather than H_1 has a concrete planning meaning. In green-dense cities the structure that matters is not isolated "deserts" surrounded by served land but the connectivity of the served regions themselves, and that connectivity is what disruption erodes.

For design, this suggests a *hypothesis* rather than a prescription: that *path redundancy*, not raw proximity, may be what protects access to nature under stress, since the descriptors associated with resilience (circuitous routing, longer segments) are hallmarks of networks offering several comparable ways to reach the same green space, so that losing one link rarely isolates a served region. If it holds, it would reframe an SDG 11.7 design brief from "put a park within x metres" toward "ensure the walk to it survives the loss of individual streets." This stops short of a planning prescription: the association is observational, the outcome is a proposed topological indicator rather than a validated resilience measure, and green-space quality, public accessibility and real hazard exposure are not modelled, so any design implication is a direction for testing, not a warrant to redesign street networks.

Limitations. The caveats are grouped by the kind of claim they qualify.

Inference and spatial dependence. The design is cross-sectional and observational, so the regression identifies *associations* between morphology and topological resilience, not causal effects. Districts are also not independent: they nest within eight cities and are spatially contiguous. City fixed effects absorb additive between-city differences, but two residual problems remain. First, eight cities furnish only eight clusters for cross-city inference; clustering the standard errors by city widens the intervals, but all three headline effects still remain significant under this conservative treatment (circuity $p = 2 \times 10^{-4}$, mean street length $p = 5 \times 10^{-19}$, orientation entropy $p = 0.036$; Table 3). Eight clusters is nonetheless modest, so greater weight is placed on the estimated spatial models rather than on small-cluster asymptotics. Second, the residuals retain weak positive spatial autocorrelation within most cities (global Moran's I up to 0.18; pooled 0.24, $p < 10^{-100}$). This is addressed by estimating spatial-error and spatial-lag models on a within-city weights matrix (Table 3): both absorb strong spatial dependence ($\hat{\lambda} = 0.53$, $\hat{\rho} = 0.51$), yet all three headline descriptors keep their sign and remain significant at $|z| \geq 5$. The morphology signal is therefore not an artefact of spatial autocorrelation. Broader replication—more cities, and cities with sparser OpenStreetMap coverage than this set—remains the natural next step.

Omitted variables. Morphology is only one of many district attributes that could be associated with resilience. The four that the data support are controlled—green-space area, road hierarchy, relief and city-centre distance (Table 5)— and the headline morphology effects survive their inclusion, so the association is not explained away by these four (which is weaker than ruling them out as causes). But population and built density, land-use mix, park-level size and quality, socioeconomic status and pedestrian-infrastructure quality would require external data (e.g. census and GHSL built-up layers) and remain uncontrolled; city fixed effects remove their between-city component but not their within-city variation, so some residual omitted-variable bias may persist.

Shared origin of predictor and outcome. Both the morphology descriptors and the resilience outcome are computed from the *same* street network, so part of the association is structural rather than a clean independent-variable/dependent- variable relationship: a district cannot have, say, low circuity and a routing topology that behaves as if it were highly circuitous. The regression is therefore read as characterising *which* formal properties co-vary with resilient access, not as an external cause acting on an independent outcome. Two observations temper the concern: the morphology descriptors explain only a modest share of the variance ($R^2 \approx 0.20$), so the outcome is far from a deterministic re-encoding of the predictors; and the accessibility field also depends on the green-space configuration, which is exogenous to street form. Still, a design that predicted one city's resilience from another's morphology would be needed to break the shared-origin coupling entirely.

What the metric measures, and how far it is validated. $D(\rho)$ is a *proposed proxy* for access resilience, and this is the study's central limitation. What is established rigorously is an *association* between morphology and this particular Wasserstein/AUC topological indicator; that circuity and mean street length "explain resilience" in a broader sense follows only to the extent that the indicator is itself a valid measure of resilience, which is not fully demonstrated. Evidence of two kinds is offered. Its *construct validity* is supported by its behaviour—it rises with disruption over most of the range (though not monotonically at high ρ, where finite classes vanish), builds on field-perturbation stability of the diagram distance on the fixed domain [5] (with the domain-change caveat of Section 4), and yields a morphology relationship that survives disruption model, district resolution, denoising threshold, pivot count, spatial-error/lag modelling and four omitted-variable controls. On *criterion* validity, the benchmark against a distance-based accessibility-degradation measure is positive at the level at which the metric is used. Within city, the unit the fixed effects identify from, the topological AUC correlates $+0.36$ ($p < 10^{-79}$) with the benchmark and is positive in all five cities of the original core, while remaining non-redundant (the morphology signal survives controlling for it). That validity has a definite scope: it holds for *graded* access degradation, not for total *disconnection*. Within city, AUC correlates near zero-to-negative with connectivity degradation, because unreachable nodes leave the finite diagram, so the metric measures the resilience of the *connected* served structure and should be paired with a disconnection count for total access loss. An attempt to fold disconnection into a single scalar is reported, and the result is instructive. A finite cap ($\tau = 60$ min) recovers the connectivity sign ($-0.10 \to +0.02$ on a two-city check) but erodes the graded-distance correlation ($+0.41 \to +0.06$), and a principled extended-persistence diagram, which gives the disconnected components finite coordinates in a separate channel, behaves the same way on Amsterdam (connectivity $+0.31$ but distance only $+0.05$), because the high-persistence component classes dominate a single Wasserstein match. Graded resilience and disconnection are, then, two channels that a single scalar conflates; reporting them separately, as done here, is the right treatment. A joint multi-channel metric with a full head-to-head against an established measure (a population-weighted two-step floating catchment degradation, say, or observed post-disruption service loss) is the natural next step. Two further caveats: AUC depends on network and district size and on the number of H_0 classes, so it is comparable across cities only in relative ranking, not in absolute level (city fixed effects absorb this scale in the regression), and ρ^* depends on the chosen disruption range. The down-turn of the İstanbul and Phoenix curves at $\rho = 0.5$ reflects finite nodes dropping out of the diagram under near-total fragmentation, not recovered access, and marks the upper edge of the metric's useful range. Green spaces are treated as equivalent point sources at their centroids, ignoring size, quality and public/private status; a large park reduced to its centroid can misplace the true walking distance to its entrances, and the sensitivity of the results to multi-entrance or area-based access points is not tested here. OSM mapping completeness and semantic consistency also vary across cities and are not quantified, so data-quality heterogeneity is not fully controlled. The class-exclusion sensitivity below addresses the tag-mix concern but not these geometric and completeness ones.

Disruption and data. Random removal is spatially uniform and does not reproduce the clustered failures of a real flood or earthquake; targeted removal addresses criticality but not spatial correlation. The real 100-year flood scenario added above makes this concrete: it materially affects only Vienna and does not carry the distributed-disruption morphology signal, showing that resilience to a localised hazard is a genuinely different quantity—one this paper does not claim to explain. Earthquake and multi-hazard disruption models are left to future work. Targeted betweenness is approximated from k=500 sampled pivots, as exact $O(V|E|)$ betweenness is intractable at city scale;

the ranking is stable in the pivot count: on the Amsterdam network, halving or doubling k (to 250 or 1,000) leaves the edge ranking essentially unchanged (Spearman $\rho = 0.985$ against k=500, with 77%/90% overlap of the top-decile edges)—so the targeted scenario is not an artefact of the pivot count. Results use a persistence-thresholded H_0 metric, so reported magnitudes are denoised estimates; varying the threshold over a 4× range (0.5–2.0 walk-minutes) preserves every coefficient sign and keeps the three headline descriptors significant at $p < 10^{-6}$, and the full disruption grid confirms the signs, significances, ranking and ρ^*. OSM completeness varies by city and is not quantified here, and the network-distance Vietoris–Rips variant of the filtration is left to future work.

Modelling choices. Two inputs could in principle shape the results: the inclusive green-space definition and the constant 4.8 km/h walking speed. Both were tested. The strict, accessibility-conservative green-space subset leaves the headline signs and significance unchanged (Section 6); and, because speed enters only as a global rescaling of the field, the walking-speed choice cannot alter the coefficient signs, significances or the city ranking, which is confirmed empirically ($\rho > 0.98$ between the 4.8 and 4.0/5.0 km/h district AUCs). These are therefore not load-bearing assumptions.

Finally, "resilience" here is a *structural* property—the robustness of the access topology to synthetic edge removal, measured on a single cross-sectional snapshot per city—not a temporal recovery process; how access rebuilds over time after a real disruption, and the term should be read in that narrower, topological sense.

Read together, these caveats narrow the claim: within these eight cities, and under synthetic edge-removal disruption, street-network morphology is a consistent descriptive correlate of green-space access resilience—robust in sign across disruption models, district resolutions, spatial specifications and the added controls—rather than a causally identified or cross-city-generalisable driver.

8 Conclusion

This study defines a topological metric for the resilience of green-space access, the Wasserstein degradation curve of persistent-homology diagrams under network disruption, and showed across 3,999 districts in eight contrasting cities on four continents that street-network morphology is associated with how robustly a city's pedestrian network keeps its *connected* served locations close to nature. Two effects are robust throughout—longer street segments (the largest) and more circuitous routing are associated with greater resilience, and both survive a wild cluster bootstrap and spatial-error and spatial-lag models—while higher orientation entropy is associated with less resilience and is strong in the spatial models but less certain across cities. Two boundaries frame the contribution. The relationship is an *association*, not a causal effect: the design is cross-sectional and observational. And it is an association with the *proposed* topological indicator; whether that indicator is a fully valid measure of resilience—rather than a well-behaved, configuration-sensitive proxy—is the study's central open question. Eight cities also still furnish only eight clusters (see Limitations). The paper is precise about what the metric captures: the resilience of the *connected* access structure, validated within city against a distance-based accessibility benchmark, rather than total accessibility including outright disconnection. The mathematics of persistence turns a static SDG 11.7 indicator into a dynamic, form-aware one. Future work will extend the real-hazard analysis beyond river floods to earthquakes and multi-hazard scenarios, and add a network-metric Vietoris–Rips filtration, toward a resilience-aware accessibility standard for sustainable cities.

Reproducibility. All data are open (OpenStreetMap, GHS-UCDB) and the pipeline is open-source Python (OSMnx, NetworkX, GeoPandas, GUDHI/`persim`, H3, `statsmodels`); every reported number is recorded in the project's results log and traces to a committed artifact.

Table 3: Inference robustness of the eight-city district regression: city-clustered standard errors (eight clusters, the most conservative treatment of the nesting) and estimated spatial-error and spatial-lag models (ML, within-city H3 weights). Coefficients are on AUC. All three headline descriptors survive city clustering and both spatial models, despite substantial estimated spatial dependence.

Morphology descriptor	coef (OLS)	p (cluster)	Sp. error z	Sp. lag z
circuity	−7.39	2.2×10^{-4} ***	−5.2	−6.6
mean street length	−0.050	4.5×10^{-19} ***	−7.5	−8.8
orientation entropy	+1.85	0.036 *	+10.1	+9.6
intersection density	+0.02	0.89	−1.5	−1.2
spatial parameter			$\hat{\lambda} = 0.53$	$\hat{\rho} = 0.51$

A Robustness and sensitivity analyses

This appendix collects the diagnostics summarised in the Results. Throughout, the three headline descriptors are circuity, mean street length and orientation entropy.

Spatial dependence. Districts nest within cities and are spatially contiguous, so the fixed-effects OLS residuals are not independent. Global Moran's I on first-ring H3 adjacency is positive and significant in every city (up to 0.18 in Barcelona and Singapore, mostly permutation $p \leq 0.01$), and Anselin Lagrange-multiplier tests on the OLS residuals decisively reject the non-spatial model (LM-error $p < 10^{-135}$, LM-lag $p < 10^{-177}$; the robust LM statistics favour the lag form, 227 vs. 31), with both spatial models cutting the AIC by more than 500 relative to OLS (22,063 → 21,475 for the error model, 21,376 for the lag model), so spatial-error and spatial-lag models are estimated spatial-error and spatial-lag models (Table 3). Both absorb strong dependence ($\hat{\lambda} = 0.53$, $\hat{\rho} = 0.51$), yet the three headline descriptors keep their sign and stay strongly significant ($|z| \geq 5$). City-clustered standard errors (Table 3), the most conservative treatment of the eight-cluster design, leave all three significant, including orientation entropy.

Multicollinearity and cross-city heterogeneity. The headline descriptors are not collinear: within-city, the largest pairwise correlation among circuity, mean street length and orientation entropy is 0.31, and all variance-inflation factors are below three (the only strong pair is mean street length and intersection density, $r = -0.80$, which is why intersection density is unstable and not relied on). The pooled slopes are also not a pooling artefact: fitting the regression separately per city, each headline sign holds in seven of the eight cities (Table 4), with Nairobi the exception for circuity and Phoenix for orientation entropy.

Disruption scenarios (five-city core). Under *targeted* disruption (removal of high-betweenness edges, ranked by k=500-sampled betweenness) every coefficient sign in Table 2 is reproduced: circuity −5.77 ($p = 4.6 \times 10^{-10}$), mean street length −0.034 ($p = 2.7 \times 10^{-13}$), orientation entropy +0.50 ($p = 1.6 \times 10^{-3}$), and intersection density (not significant under the full random grid) reaching $p = 0.022$ (−0.135). Absolute AUC values are lower (Amsterdam mean AUC 2.03 vs. 4.46 under random) but the ranking is unchanged.

Table 4: Per-city regression coefficients (AUC on morphology, one city at a time). Each headline sign—circuity (−), mean street length (−), orientation entropy (+)—holds in seven of the eight cities.

City	circuity	mean street length	orientation entropy
Amsterdam	−4.07	−0.04	+1.17
Barcelona	−6.57	+0.02	+6.17
Bogotá	−18.62	−0.21	+1.46
İstanbul	−14.51	−0.03	+3.59
Nairobi	+4.86	−0.05	+2.56
Phoenix	−3.70	−0.06	−0.72
Singapore	−4.12	−0.04	+4.58
Vienna	−5.56	−0.04	+3.39

Real flood-hazard scenario. The earlier low-elevation proxy with a *real* 100-year river-flood scenario, using the open JRC Global Flood Hazard maps [7] (floodMapGL, water depth per ≈ 1 km cell): a node is flood-exposed if it lies in a cell with ≥ 0.5 m of inundation, and the hazard removal progressively cuts flood-exposed edges. Two honest findings follow. First, real 100-year exposure among these urban centres is strongly city-specific: it is material only in Vienna (23% of nodes, on the Danube floodplain) and negligible elsewhere (Phoenix 0.7%, Amsterdam 0.4%, Barcelona 0.3%; İstanbul, Bogotá, Singapore and Nairobi essentially 0%, their centres sitting off the modelled river-flood plain). Second, where the scenario bites (Vienna), the morphology signal established for *distributed* disruption does *not* cleanly carry over to this *spatially clustered* flood: in a pooled flood-scenario regression the morphology coefficients are non-significant, and within Vienna only circuity keeps a weak same-signed effect (Spearman −0.13, $p = 0.006$) while orientation entropy even reverses. This is consistent with, and concretely demonstrates, the caveat that resilience to network-wide disruption is a different quantity from resilience to a localised, geographically concentrated hazard; morphology governs the former, not necessarily the latter.

Disruption-grid and resolution sensitivity (five-city core). A coarser 5×3 disruption grid leaves every sign unchanged and keeps the three headline descriptors significant at $p < 10^{-3}$ (circuity −8.99, mean street length −0.0478, orientation entropy +1.17), with the city AUC ranking and ρ^* values within ≈ 0.02 of the full 8×10 grid. The one descriptor that moves is intersection density: significant on the coarse grid (−0.199, $p = 8.6 \times 10^{-3}$) but not on the full grid ($p = 0.22$). Re-tiling at H3 resolutions 7 (485 districts) and 9 (13,782 districts) likewise leaves the three headline descriptors significant at $p < 10^{-3}$ with constant signs across a 28× range of district counts; magnitudes scale with cell size but no sign flips. Figure 5 summarises both axes.

Persistence-threshold sensitivity (five-city core). Varying the H_0 denoising threshold over a 4× range leaves the headline unchanged: at 0.5, 1.0 and 2.0 walk-minutes the three descriptors keep their sign and stay significant at $p < 10^{-6}$ (circuity $-11.20/-10.16/-8.03$; mean street length $-0.050/-0.048/-0.042$; orientation entropy $+1.10/+1.01/+0.93$). Magnitudes shrink smoothly as more denoising is applied but never change sign, and intersection density stays weak (ns at 0.5 and 1.0, $p = 0.04$ at 2.0). The betweenness pivot count is equally inconsequential: on the Amsterdam network, halving or doubling k (to 250 or 1,000) leaves the edge ranking essentially unchanged (Spearman $\rho = 0.985$ against k=500, with 77%/90% overlap of the top-decile edges).

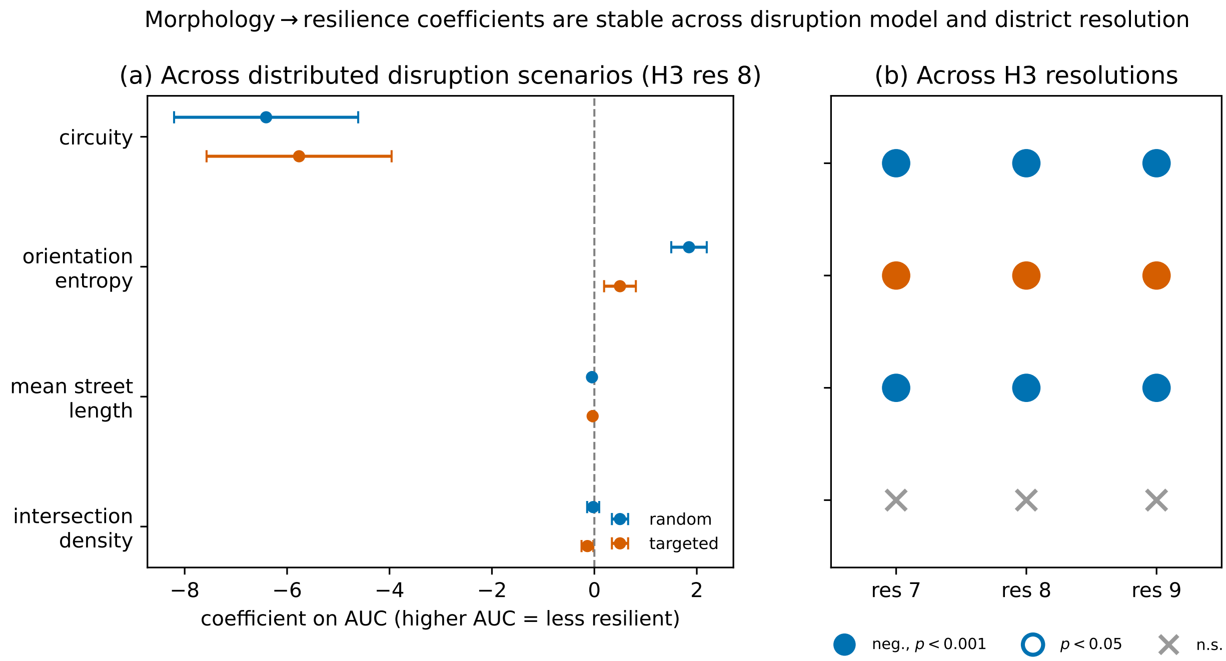

Figure 5: Robustness of the morphology↔resilience relationship. **(a)** Regression coefficients (with 95% confidence intervals) for each morphology descriptor under the two distributed disruption scenarios (random and targeted) at H3 resolution 8; the coefficient signs agree, and the three headline descriptors (circuity, mean street length, orientation entropy) stay significant, while intersection density crosses zero under random disruption. **(b)** Sign and significance of each coefficient across H3 resolutions 7/8/9: the three headline descriptors stay significant with a constant sign, while intersection density, the weakest and grid-sensitive effect, is not significant at the headline full-grid specification.

Omitted-variable controls (five-city core). City fixed effects absorb only between-city confounds, so the district-level controls that the data support are added—total green-space area, the share of edges on the trunk/primary/secondary/tertiary road hierarchy, within-district relief (elevation standard deviation), and distance to the urban-centre centroid—and refit (Table 5). Three of the four controls are themselves significant (more green area and greater centre distance with greater resilience, more major-road edges with less), yet the three headline descriptors keep their sign and stay significant at $p < 10^{-11}$ (circuity -9.55, mean street length -0.039, orientation entropy $+1.11$). Population and built density, land-use mix, socioeconomic status and pedestrian-infrastructure quality require external data and remain uncontrolled (Limitations).

Declarations

Funding. The author received no specific funding for this work.

Competing interests. The author declares no competing interests.

Data availability. All input data are open and publicly available. Pedestrian networks and green spaces were obtained from OpenStreetMap via OSMnx; urban-centre boundaries from the GHS

Table 5: Omitted-variable robustness (five-city core): district regression before and after adding four controls, $n = 2{,}603$, with city fixed effects. The three headline morphology effects keep their sign and significance; the added controls explain additional variance (R^2 0.17 → 0.21). Sig.: $^{***}p < 0.001$, $^{**}p < 0.01$.

Variable	base coef (p)	+controls coef (p)
circuity	−10.16 ***	−9.55 ***
mean street length	−0.048 ***	−0.039 ***
orientation entropy	+1.01 ***	+1.11 ***
intersection density	−0.09	−0.10
green-space area (km^2)	—	−2.25 **
major-road share	—	+4.21 ***
relief (m)	—	−0.008
centre distance (km)	—	−0.107 ***

Urban Centre Database (GHS-UCDB R2019A, European Commission Joint Research Centre); residential population from the GHS-POP grid (R2023A, 1 km) for the population-weighted validation; 100-year river-flood depth from the open JRC Global Flood Hazard maps (floodMapGL) for the real-flood scenario; and elevation data from the public AWS Terrain Tiles service. No new data were collected. The derived artifacts underlying every reported figure and table (per-district resilience and morphology tables, regression outputs, and city-level resilience curves) are recorded in the project's results log, released with the code at `https://github.com/navigaai/nnj-sdg11-topology`.

Code availability. The complete analysis pipeline is implemented in open-source Python (OSMnx, NetworkX, GeoPandas, GUDHI, `persim`, H3, `statsmodels`) and is publicly available at `https://github.com/navigaai/nnj-sdg11-topology`, together with the configuration files and the results log that reproduce every reported number.

Author contributions. S. Emekci (ORCID 0000-0002-5470-6485) conceived the study, developed the method and software, performed the analysis, and wrote the manuscript.

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